

IoT and its shortcomings

As we all know, IoT stands for the Internet of Things. Since we humans have been using the Internet for a long time to communicate with each other, the term IoT is also referred to as the Internet of People. The word ‘things’ generally refers to physical objects such as a lightbulb, refrigerator, or a machine in a factory. Anything that can communicate with other entities via the Internet can be a part of the Internet of Things.

IoT comprises sensors, actuators, and controllers. Sensors are electromechanical or electrochemical devices, which sense physical entities and convert them into electrical signals.

Actuators are devices that receive those electrical signals and perform a physical action. And finally, the controller, which acts as the brain, controls both sensors and actuators. It provides electrical signals to the actuator and reads these signals from the sensor.

When IoT devices communicate information in conjunction, the data gets stored in the cloud. IoT architecture may look simple but it has several drawbacks. Absence of the Internet connection, for instance, can lead to catastrophic failures as the sensor and the actuator won't be able to communicate with the cloud, causing data loss.

Also, if every individual sensor tries to communicate with the cloud at the same time, the transmission of many packets will interfere or slow down the path from the sensor or controller to the cloud. To avoid such scenarios, edge computing is used.

When edge computing and IoT work together, the sensor and controller do not talk to the cloud directly. Instead, communication takes place through an edge gateway, which aggregates all the data and transfers it to the cloud. Here, no Internet connection is required between the sensor/actuator and the controller and the edge gateway.

This way, response is obtained even when there is no Internet connection. This can reduce the latency and the number of

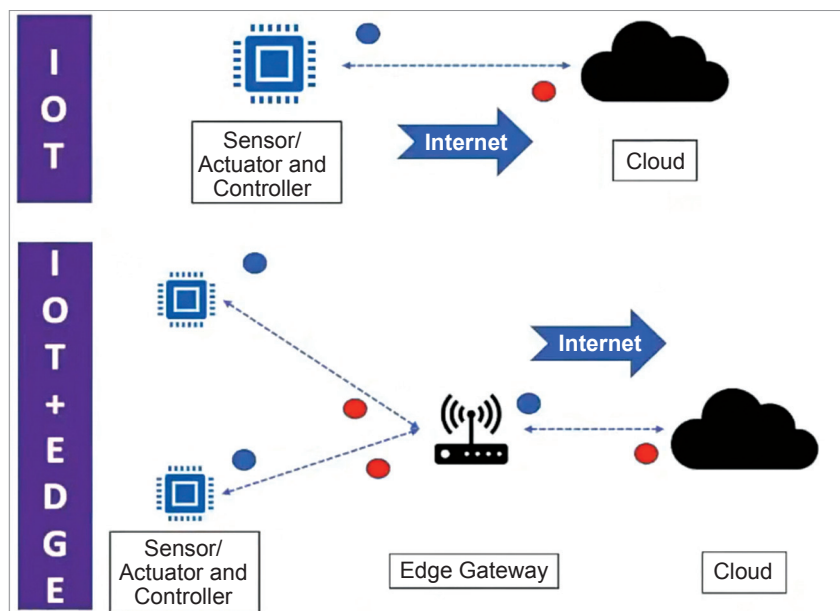


Figure 2: IoT in a nutshell

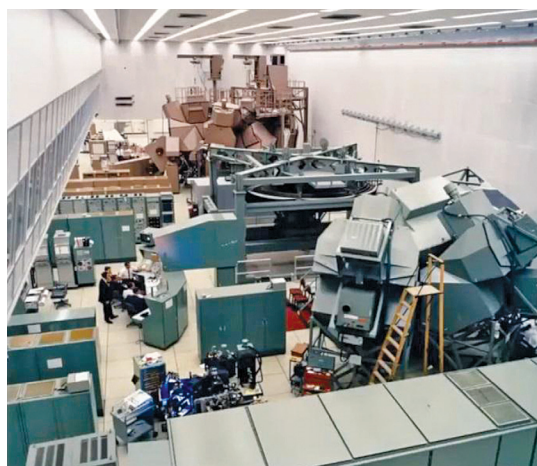


Figure 3: Apollo simulators at mission control in Houston (Credit: NASA)

messages being sent to the cloud. The edge gateway can process data locally and send only important and meaningful information to the cloud.

Where Digital Twin meets IoT

Digital Twin without IoT is nothing but a static simulation model. With the addition of Digital Twin we can get real-time data from physical things via IoT. That makes the entire thing a living-learning model.

Let us understand this from the perspective of a car. To simulate a car in a difficult situation and examine how

it will behave, we need to feed in some parameters. But the actual result is sometimes different from that obtained from the simulation. Instead, if we connect particular sensors, which in turn are getting directly displayed over the Digital Twin, then that will give a whole lot more realistic data.

Where Digital Twin meets AI

Digital Twin is a continuously learning

system, powered by various algorithms.

Let us assume that we have several pumps in our factory. To make Digital Twin models of each pump, we first need to make 3D models of them. The engineer will work to collect the physical, mechanical, electrical, and other parameters to make a 3D design. And after that the actual objects will be fitted with sensors, which will deliver data to the digital replicas. And this Twin will display the live data and store it in a database (as historical data). This data can be used to train a model.